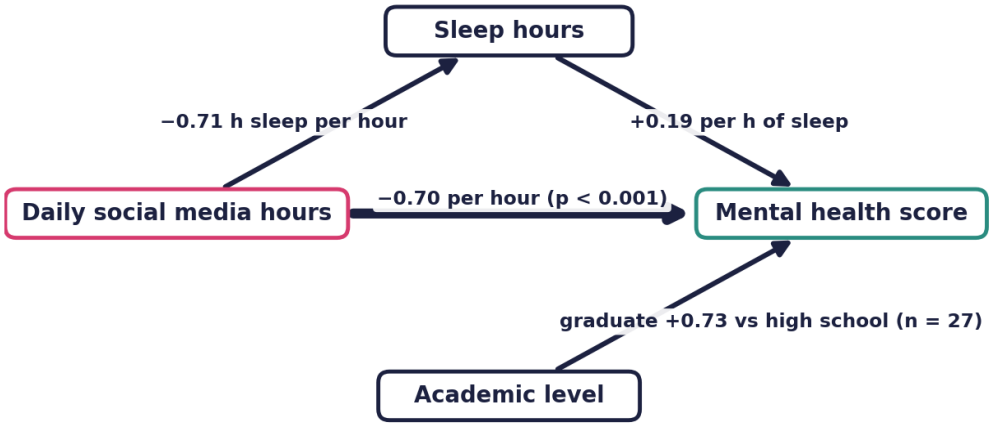
Three System Maps Comparison Report

GROUP32, ACDT 2026 Semester 2, Project 1: Does scrolling more mean feeling worse? | Website: dhkim060225-web.github.io/group32-sns-mental-health | Repository: github.com/dhkim060225-web/group32-sns-mental-health

Why three maps

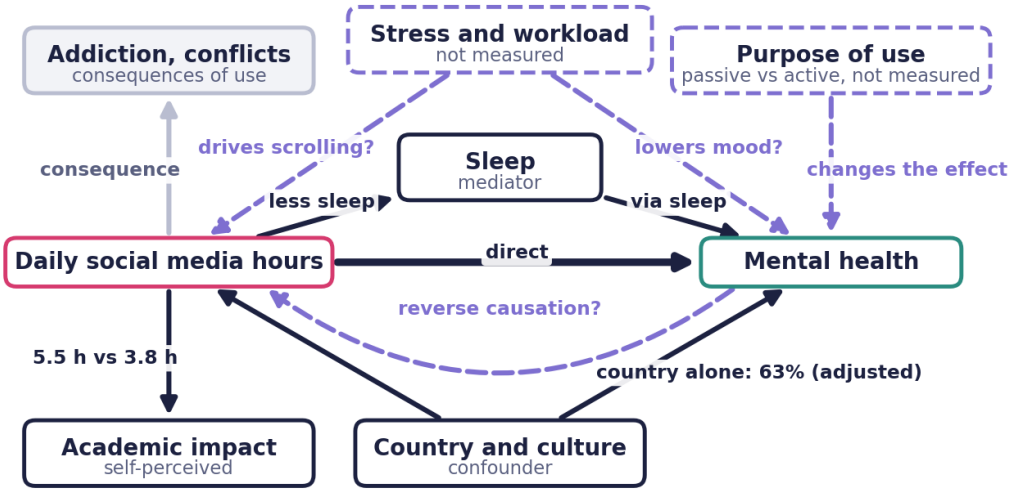
The same question (is daily social media use linked to lower student mental health?) can be drawn three ways. Map A keeps only the links our regressions support at $p < 0.05$. Map B adds what theory and our discussion say should be there, including factors we did not measure. Map C shows what remains after double machine learning (DML) removes measured background differences. Each map is a hypothesis about structure, not proof of cause.

Map A: statistical significance (from regression results)



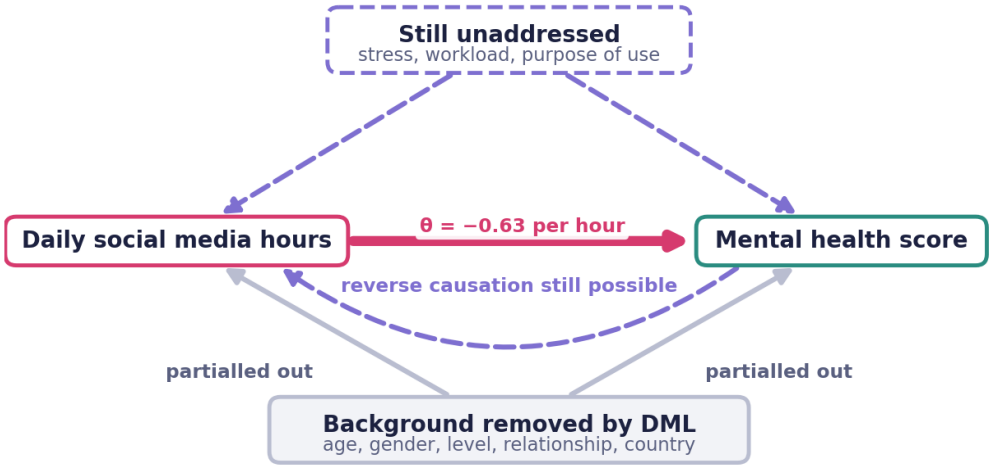
Built from our original hypothesis H_1: hours are linked to mental health score. Estimated slope −0.70 per hour (95% CI −0.74 to −0.66, $t = -35.5$), $R^2 = 0.64$. Sleep carries about 20% of the association. Graduate students score 0.73 points higher than high school students ($p < 0.001$; only 27 high school students), but academic level does not change the hours slope ($p = 0.81$).

Map B: theory and scenarios (including hidden features)



Dashed lavender links are hypotheses that our data cannot test. Stress, workload and purpose of use were not measured. Reverse causation (low mood leading to more scrolling) cannot be ruled out in a cross-sectional survey. The sleep pathway is consistent with research on screen time and sleep in children and adolescents, which does not confirm causation (Hale & Guan, 2015). Country is the strongest measured confounder: alone it explains 63% of score variation after adjustment, about as much as hours (64%).

Map C: machine-learning approach (double machine learning)



Random forests remove what the background variables predict about both hours and mental health, using 5-fold cross-fitting. The remaining hours effect is −0.63 per hour (95% CI −0.68 to −0.59), smaller than the raw −0.70 but still clearly negative. The interval assumes independent students, so with country clustering the true uncertainty is wider. The method adjusts only for measured variables, so the unmeasured factors in Map B remain open.

Comparison of the three maps

	Map A: statistical	Map B: theory and scenarios	Map C: after double ML
Built from	Regression results, $p < 0.05$	Our discussion, research on sleep and wellbeing, data audit	Random forests with cross-fitting
Hours effect	−0.70 per hour	Direction and size not testable here	−0.63 per hour
Adds	A clear, quantified, testable link	Stress, purpose of use, reverse causation, culture	Measured confounders removed without assuming a straight line
Misses	Anything unmeasured; treats association as if it had a direction	Most links cannot be tested in this data	Unmeasured confounders and reverse causation
Policy if used alone	Cut screen time for everyone	Target stress, sleep and type of use	Hours are a real but partial warning sign

Justification 1: Why certain non-significant variables might still matter

Non-significant is not the same as no effect. The hours slope does not differ significantly by gender ($p = 0.31$) or by academic level ($p = 0.81$). Yet the high school group has only 27 students, so the test can detect only large differences. A non-significant result here means we have no evidence of a difference, not that the groups are the same.

Platform type matters, and it matters for the policy. Raw slopes range from −0.41 (WhatsApp) to −1.05 (Twitter); a common slope is rejected ($F(8, 650) = 3.59$, $p < 0.001$). Country could explain part of this, so we also compared students from the same country on the same platform. For the four platforms where that is possible the slopes still differ (Facebook −0.74, Twitter −0.91, Instagram −0.36, TikTok −0.31; $F(3, 432) = 13.0$, $p < 0.0001$). WhatsApp's slope cannot be checked this way: all 54 users live in four countries and give one score per country, so its raw slope reflects differences between countries only. Platform is only a proxy for purpose of use, so these results show that the association differs by platform, not that passive and active use affect wellbeing differently.

Variables we never measured still matter. Stress and workload are plausible causes of both heavy scrolling and low mood. If they drive both, every estimate here overstates what reducing screen time would achieve. A sensitivity analysis (Cinelli & Hazlett, 2020) shows such a confounder would need to explain at least 71% of the remaining variation in both hours and scores to erase the simple-regression slope, and 48% to halve it. Absence of data is not evidence of absence of an effect.

Justification 2: Policy implications if we ignore these hidden features

- **Ignoring country** would treat between-country differences as individual effects. Within countries the slope is −0.58 (country-clustered 95% CI −0.73 to −0.42), and for the pooled slope the clustered interval is about 2.4 times wider than the textbook one. A policy built on the pooled number would look more certain than the data allows.
- **Ignoring sleep** would miss a pathway carrying about 20% of the association and a practical place to intervene. Counsellors could ask about sleep early in a conversation.

- **Ignoring platform type** would apply one screen-time rule to every platform, even though their slopes differ. Because platform only stands in for purpose of use, the next step is to measure purpose directly before designing platform-specific advice.
- **Ignoring stress and purpose of use** would leave screen-time limits as the only lever. If stress drives heavy use, limits alone would miss the students who need support most.
- **Controlling for addiction and conflicts** would make the effect disappear (slope −0.06). These are probably consequences of heavy use, so adjusting for them hides the pattern rather than revealing a truer one.

References: Cinelli, C., & Hazlett, C. (2020). Making sense of sensitivity: Extending omitted variable bias. Journal of the Royal Statistical Society: Series B, 82(1), 39–67. Hale, L., & Guan, S. (2015). Screen time and sleep among school-aged children and adolescents: A systematic literature review. Sleep Medicine Reviews, 21, 50–58.

AI assistance: An AI assistant (Claude, by Anthropic) was used to help generate the system map diagrams in this report. The team defined the research question, chose the variables and methods, and reviewed all interpretations.